

[Home](#) / [A](#) /

Mono2Micro: A Practical and Effective Tool for Decomposing Monolithic Java Applications to Microservices

Who

Anup K. Kalia (<https://2021.esec-fse.org/profile/anupkkalia>), Jin Xiao (<https://2021.esec-fse.org/profile/jinxiao>), Rahul Krishna (<https://2021.esec-fse.org/profile/rahulkrishna>), Saurabh Sinha (<https://2021.esec-fse.org/profile/saurabhsinha>), Maja Vukovic (<https://2021.esec-fse.org/profile/majavukovic>), Debasish Banerjee (<https://2021.esec-fse.org/profile/debasishbanerjee>)

Track

ESEC/FSE 2021 Industry Papers (<https://2021.esec-fse.org/track/fse-2021-industry>)

When

Fri 27 Aug 2021 17:10 - 17:20 - Architectures & Design—Cloud Computing 2 (<https://2021.esec-fse.org/track/fse-2021-papers#program>) Chair(s): Luciano Baresi (<https://2021.esec-fse.org/profile/lucianobaresi>), Yu Kang (<https://2021.esec-fse.org/profile/yukang>)

🕒 **Sat 28 Aug 2021 05:10 - 05:20** - Architectures & Design—Cloud Computing 2 (<https://2021.esec-fse.org/track/fse-2021-papers#program>) Chair(s): Yu Kang (<https://2021.esec-fse.org/profile/yukang>)

Abstract

In migrating production workloads to cloud, enterprises often face the daunting task of evolving monolithic applications toward a microservice architecture. At IBM, we developed a tool called Mono2Micro to assist with this challenging task. Mono2Micro performs spatio-temporal decomposition, leveraging well-defined business use cases and runtime call relations to create functionally cohesive partitioning of application classes. Our preliminary evaluation of Mono2Micro showed promising results.

How well does Mono2Micro perform against other decomposition techniques, and how do practitioners perceive the tool? This paper describes the technical foundations of Mono2Micro and presents results to answer these two questions. To answer the first question, we evaluated Mono2Micro against four existing techniques on a set of open-source and proprietary Java applications and using different metrics to assess the quality of decomposition and tool's efficiency. Our results show that Mono2Micro significantly outperforms state-of-the-art baselines in specific metrics well-defined for the problem domain. To answer the second question, we conducted a survey of twenty-one practitioners in various industry roles who have used Mono2Micro. This study highlights several benefits of the tool, interesting practitioner perceptions, and scope for further improvements. Overall, these results show that Mono2Micro can provide a valuable aid to practitioners in creating functionally cohesive and explainable microservice decompositions.

DOI

🔗 <https://doi.org/10.1145/3468264.3473915> (<https://doi.org/10.1145/3468264.3473915>)

Anup K. · Kalia

**IBM Research**

United States

<https://2021.esec-fse.org/profile/anupkalia>**Jin Xiao****IBM Research**

United States

<https://2021.esec-fse.org/profile/jinxiao>**Rahul Krishna****IBM Research**

United States

<https://2021.esec-fse.org/profile/rahulkrishna>**Saurabh Sinha****IBM Research**

United States

<https://2021.esec-fse.org/profile/saurabhsinha>**Maja Vukovic****IBM Research**

United States

<https://2021.esec-fse.org/profile/majavukovic>**Debasish Banerjee****IBM**

n.n.

<https://2021.esec-fse.org/profile/debasishbanerjee>

Session Program

Fri 27 Aug

Displayed time zone: **Athens** [change](#)

17:00 - 18:00 **Architectures & Design—Cloud Computing 2**  +12h

Industry Papers (<https://2021.esec-fse.org/track/fse-2021-industry>) / **Research Papers** (<https://2021.esec-fse.org/track/fse-2021-papers>)

Chair(s): Luciano Baresi (<https://2021.esec-fse.org/profile/lucianobaresi>) Politecnico di Milano, Yu Kang (<https://2021.esec-fse.org/profile/yukang>) Microsoft Research, Beijing, China

17:00 **10m** ☆ **Onion: Identifying Incident-Indicating Logs for Cloud Systems**

Paper

Industry Papers

Xu Zhang (<https://2021.esec-fse.org/profile/xuzhang>) Microsoft Research, Yong Xu (<https://2021.esec-fse.org/profile/yongxu>) Microsoft Research, Si Qin (<https://2021.esec-fse.org/profile/siqin>) Microsoft Research, Shilin He (<https://2021.esec-fse.org/profile/shilinh1>) Microsoft Research, Bo Qiao (<https://2021.esec-fse.org/profile/boqiao>) Microsoft Research, Ze Li (<https://2021.esec-fse.org/profile/zeli>) Microsoft Azure, Hongyu Zhang (<https://2021.esec-fse.org/profile/hongyuzhang>) University of Newcastle, Xukun Li (<https://2021.esec-fse.org/profile/xukunli>) Microsoft Azure, Yingnong Dang (<https://2021.esec-fse.org/profile/yingnongdang>) Microsoft Azure, Qingwei Lin (<https://2021.esec-fse.org/profile/qingweilin>) Microsoft Research, Murali Chintalapati (<https://2021.esec-fse.org/profile/muralichintalapati>) Microsoft Azure, Saravanakumar Rajmohan (<https://2021.esec-fse.org/profile/saravanakumarrajmohan>) Microsoft 365, Dongmei Zhang (<https://2021.esec-fse.org/profile/dongmeizhang>) Microsoft Research

 DOI (<https://doi.org/10.1145/3468264.3473919>)

17:10 **10m** ☆ **Mono2Micro: A Practical and Effective Tool for Decomposing Monolithic Java Applications to Microservices**

Paper

Industry Papers

Anup K. Kalia (<https://2021.esec-fse.org/profile/anupkalia>) IBM Research, Jin Xiao (<https://2021.esec-fse.org/profile/jinxiao>) IBM Research, Rahul Krishna (<https://2021.esec-fse.org/profile/rahulkrishna>) IBM Research, Saurabh Sinha (<https://2021.esec-fse.org/profile/saurabhsinha>) IBM Research, Maja Vukovic (<https://2021.esec-fse.org/profile/majavukovic>) IBM Research, Debasish Banerjee (<https://2021.esec-fse.org/profile/debasishbanerjee>) IBM

 DOI (<https://doi.org/10.1145/3468264.3473915>)

17:20 **10m** ☆ **RAPID: Checking API Usage for the Cloud in the Cloud**

Paper

Industry Papers

Michael Emmi (<https://2021.esec-fse.org/profile/michaellemmi>) Amazon Web Services, Liana Hadarean (<https://2021.esec-fse.org/profile/lianahadarean>) Amazon Web Services, Ranjit Jhala (<https://2021.esec-fse.org/profile/ranjitjhala1>) University of California at San Diego; Amazon Web Services, Lee Pike (<https://2021.esec-fse.org/profile/leepike1>) Amazon Web Services, Nico Rosner (<https://2021.esec-fse.org/profile/nicorosner>) Amazon Web Services, Martin Schäf (<https://2021.esec-fse.org/profile/martinschaf>) Amazon Web Services, Aritra Sengupta (<https://2021.esec-fse.org/profile/aritrasengupta>) Amazon Web Services, Willem Visser (<https://2021.esec-fse.org/profile/willemvisser2>) Amazon Web Services

 DOI (<https://doi.org/10.1145/3468264.3473934>)

17:30 **30m** ☆ **Q&A (Architectures & Design—Cloud Computing 2)**

Live Q&A

Research Papers

Sat 28 Aug

Displayed time zone: **Athens** [change](#)

05:00 - 06:00  **Architectures & Design—Cloud Computing 2**

Research Papers (<https://2021.esec-fse.org/track/fse-2021-papers>) / **Industry Papers** (<https://2021.esec-fse.org/track/fse-2021-industry>)

Chair(s): Yu Kang (<https://2021.esec-fse.org/profile/yukang>) Microsoft Research, Beijing, China

05:00 10m ☆ Onion: Identifying Incident-Indicating Logs for Cloud Systems

Paper

Industry Papers

Xu Zhang (<https://2021.esec-fse.org/profile/xuzhang>) Microsoft Research, Yong Xu (<https://2021.esec-fse.org/profile/yongxu>) Microsoft Research, Si Qin (<https://2021.esec-fse.org/profile/siqin>) Microsoft Research, Shilin He (<https://2021.esec-fse.org/profile/shilinhe1>) Microsoft Research, Bo Qiao (<https://2021.esec-fse.org/profile/boqiao>) Microsoft Research, Ze Li (<https://2021.esec-fse.org/profile/zeli>) Microsoft Azure, Hongyu Zhang (<https://2021.esec-fse.org/profile/hongyuzhang>) University of Newcastle, Xukun Li (<https://2021.esec-fse.org/profile/xukunli>) Microsoft Azure, Yingnong Dang (<https://2021.esec-fse.org/profile/yingnongdang>) Microsoft Azure, Qingwei Lin (<https://2021.esec-fse.org/profile/qingweilin>) Microsoft Research, Murali Chintalapati (<https://2021.esec-fse.org/profile/muralichintalapati>) Microsoft Azure, Saravanakumar Rajmohan (<https://2021.esec-fse.org/profile/saravanakumarrajmohan>) Microsoft 365, Dongmei Zhang (<https://2021.esec-fse.org/profile/dongmeizhang>) Microsoft Research

DOI (<https://doi.org/10.1145/3468264.3473919>)

05:10 10m ☆ Mono2Micro: A Practical and Effective Tool for Decomposing Monolithic Java Applications to Microservices

Paper

Industry Papers

Anup K. Kalia (<https://2021.esec-fse.org/profile/anupkalia>) IBM Research, Jin Xiao (<https://2021.esec-fse.org/profile/jinxiao>) IBM Research, Rahul Krishna (<https://2021.esec-fse.org/profile/rahulkrishna>) IBM Research, Saurabh Sinha (<https://2021.esec-fse.org/profile/saurabhsinha>) IBM Research, Maja Vukovic (<https://2021.esec-fse.org/profile/majavukovic>) IBM Research, Debasish Banerjee (<https://2021.esec-fse.org/profile/debasishbanerjee>) IBM

DOI (<https://doi.org/10.1145/3468264.3473915>)

05:20 10m ☆ RAPID: Checking API Usage for the Cloud in the Cloud

Paper

Industry Papers

Michael Emmi (<https://2021.esec-fse.org/profile/michaelemmi>) Amazon Web Services, Liana Hadarean (<https://2021.esec-fse.org/profile/lianahadarean>) Amazon Web Services, Ranjit Jhala (<https://2021.esec-fse.org/profile/ranjitjhala1>) University of California at San Diego; Amazon Web Services, Lee Pike (<https://2021.esec-fse.org/profile/leepike1>) Amazon Web Services, Nico Rosner (<https://2021.esec-fse.org/profile/nicorosner>) Amazon Web Services, Martin Schäfer (<https://2021.esec-fse.org/profile/martinschaf>) Amazon Web Services, Aritra Sengupta (<https://2021.esec-fse.org/profile/aritrasengupta>) Amazon Web Services, Willem Visser (<https://2021.esec-fse.org/profile/willemvisser2>) Amazon Web Services

DOI (<https://doi.org/10.1145/3468264.3473934>)

05:30 30m ☆ Q&A (Architectures & Design—Cloud Computing 2)

Live Q&A

Research Papers